

Project guidelines

The assessment for this module consists of a consultancy project, where you will work in project teams to investigate a real business case with one of our top industry partners. The aim is to provide analytical insight and feedback to the stakeholders on how to improve key performance indicators (KPIs).

The project consists of three parts: The report, your video presentation and your peer evaluations. Your presentation should be designed for the CEO/board of directors with low interest in technical details. However, the project report should be designed for technical experts and the data team who will read it in detail.

Project structure

Report

Your project report should be delivered as a **Jupyter Notebook**. You should use **Python** to analyse your data and implement any models.

Use the following checklist to structure your report:

- ☐ **Title page**
 - An accurate and informative title of your research
- ☐ **Summary**
 - One paragraph summarising the objective, procedure, results, discussion and conclusion of the analysis.
- ☐ **Introduction**
 - An overview of the context for the analysis, including a clear statement of the business problem
 - clear statement of your hypotheses and predictions
 - A brief overview of the approach used in your analysis.
- ☐ **Technologies and techniques**
 - An exact process description used in the analysis
 - Sufficient detail of the data analysis for someone to replicate the results (this should be automatic in Jupyter Notebooks).
- ☐ **Results**
 - A presentation and summary of the data using means, histograms and other descriptive statistics
 - Figures, tables and graphs, insofar as they help to enhance the clarity of the information (include captions)
 - An explanation of any figures, tables or graphs.
- ☐ **Discussion**
 - A short paragraph summarising the overall trend of the findings
 - A clear statement of whether your hypothesis is correct (include caveats where

necessary)

- An explanation of any errors
- A discussion of the strengths and limitations of the analysis.

☐ **Limitations and Future Directions**

☐ **References****

Accurate documentation of sources (following convention)

☐ **Technical Appendices**

Please note that your technical appendices should be well organised and be easy to replicate by the clients' data analytics team. Remember your goal is to help the client in the longer term.

Presentation

Alongside your report, you will deliver an in-person presentation to the teaching team and your cohort. The presentation should be no longer than 10 minutes (with 8 minutes being the typical target). Your audience will be the CEO and board of directors, who typically have low interest in technical details. Therefore, your presentation should focus primarily on the Introduction, Results, and Discussion sections of your report. You may choose to present as a group or nominate one individual to deliver the presentation on behalf of your team. However, all group members are required to remain on stage together, particularly contribute to the Q&A portion of the presentation.

Evaluation

On presentation day, you will also take part in a role-playing exercise by evaluating one assigned presentation from the perspective of a client. You will act as the board of directors working with a team of consultants and engage in a Q&A session with your cohort after their presentation. This activity encourages you to think critically from the client's point of view, focusing on clarity, relevance, and how effectively the presentation addresses the client's needs and objectives. You will be assessed on the quality of the questions you ask, the conversations you engage in during the Q&A session, and your submitted peer evaluation.

Grading

The purpose of this mini-consultancy project is to plan, analyse, execute and deliver actionable insights. Please visit the assessment page of the Hub for the breakdown of marks.

You will be assessed on the following competencies:

- The formulation and validation of your hypothesis
- The quality of your technical analysis
- Your storytelling and communication skills
- Your ability to convey actionable insights and feedback to the client
- Your ability to reflect and provide constructive feedback.

Project details

To succeed, you need to work together in a collaborative, efficient and systematic way, with clearly defined objectives and roles.

Project management

For this module in particular, project management skills are critical to your success. We will be providing tips and advice on project management throughout the module, but we urge you to pay attention and prioritise the following:

- Assign a project lead (per project) who will be in charge of communications, documenting progress and writing the reflective journal. The project lead will also be responsible for submitting materials on behalf of the group.
- Schedule regular meetings - at least once per week (for planning, implementation and evaluation). These formal meetings should be underpinned by daily stand-ups (see session for more details).
- Use a tool to track your progress and prioritise tasks. All group members should be allocated tasks and if someone finishes a task sooner than expected, that person should move to the next high-priority task.
- Escalate any major blockers to key stakeholders.
- Be disciplined about backing-up your work - if you are familiar with version control software (e.g. Git) use it.
- Assign someone to be in charge of QA - it's important to review your materials prior to submission. **DO NOT LEAVE THIS TO THE LAST MINUTE.**
- Make sure to rotate responsibilities throughout the duration of this module.

Project coaching

The module leader and wider teaching team will play the role of project stakeholders. We will not provide specific information on how to achieve something, but rather advise and coach you on what to try and what to prioritise.

The teaching team will offer support through weekly group coaching sessions that simulate the client-consultant relationship, mirroring real client interactions. In these meetings, the teaching team will be a representative of the clients, and the communication will be based on a typical client-consultant relationship. Keep an eye on the schedule of the teaching team via Hub and book your meetings for each week.

Collaboration

As in the workplace, your individual success lies with the success of your team. Everyone in your team must have a specific role and contribute fairly. It is your joint responsibility to ensure that you all personally understand the content of your assignments and are able to answer any questions.

As participants in a postgraduate programme, it is acceptable for groups to collaborate in figuring out technical challenges to deliver their analysis, but plagiarism will not be tolerated.